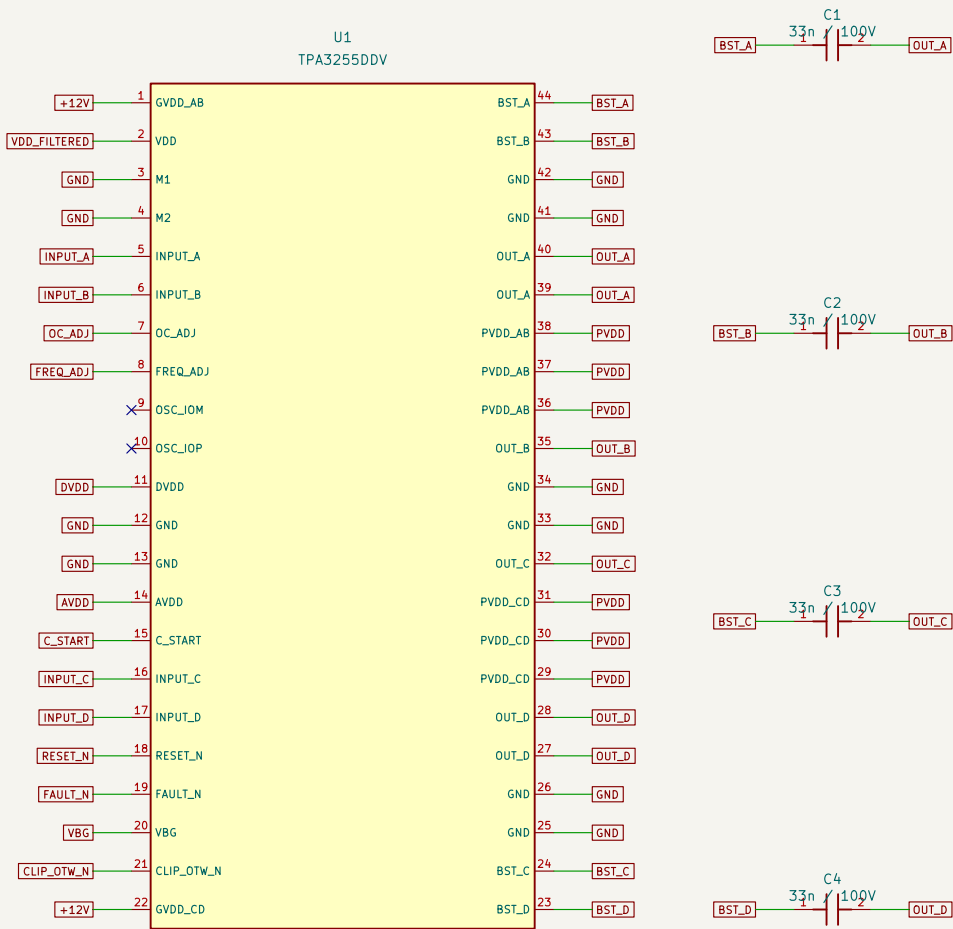


POWER STAGE / STEREO BTL / BENCH PROTOTYPE

Global net names connect all three sheets. Same labels are electrically connected.



M1=M2=0; OSC pins intentionally NC.

Top PowerPAD: ground heatsink per TI pin table.

BST and duplicate output pads use passive ERC types;

bootstrap and output ties are checked by netlist audit.

input

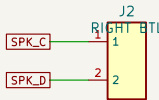
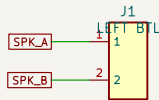
power-control

File: input.kicad\_sch

File: power-control.kicad\_sch

reset

File: reset.kicad\_sch



Speaker terminals float: never join B or D to GND.

LC values are a trial baseline; 8-ohm response needs review.

Electrical draft – no PCB / footprints / hardware validation

Sheet: /  
File: tpa3255-v02.kicad\_sch

**Title: TPA3255 V0.2 / tpa3255-v02**

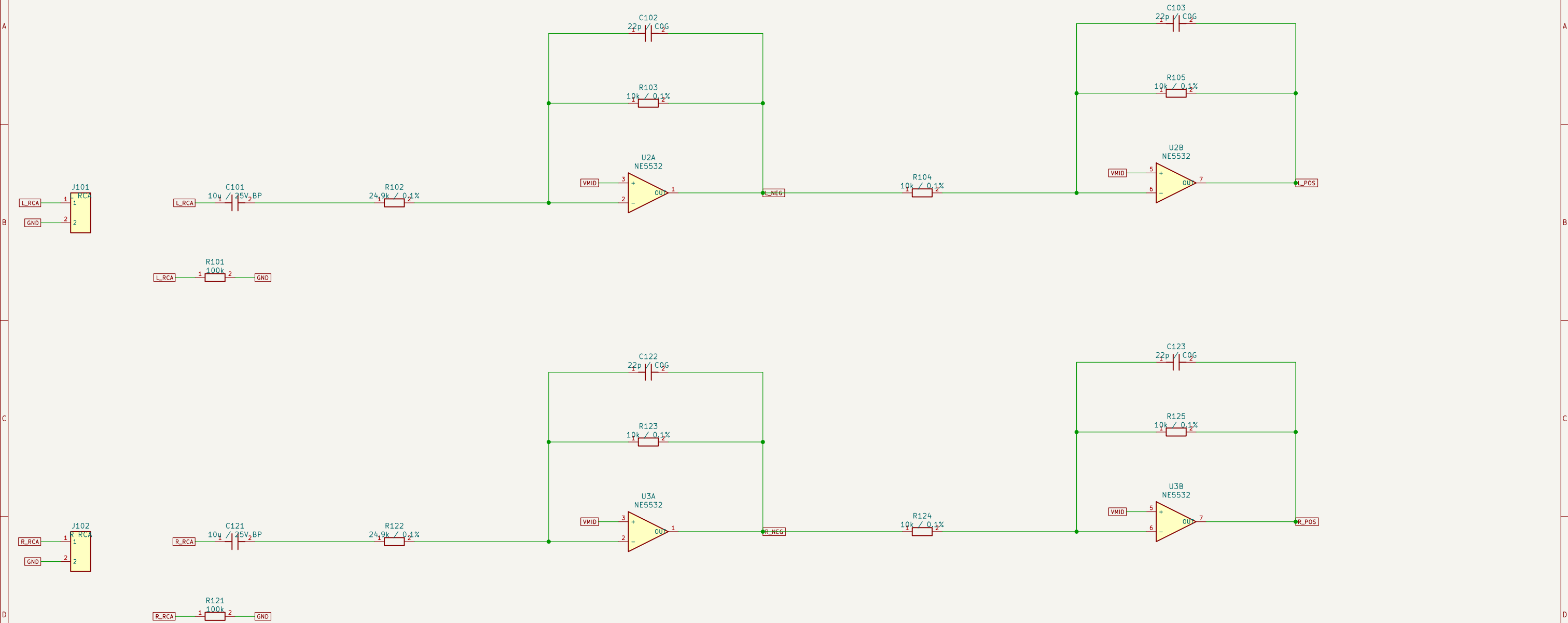
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-10

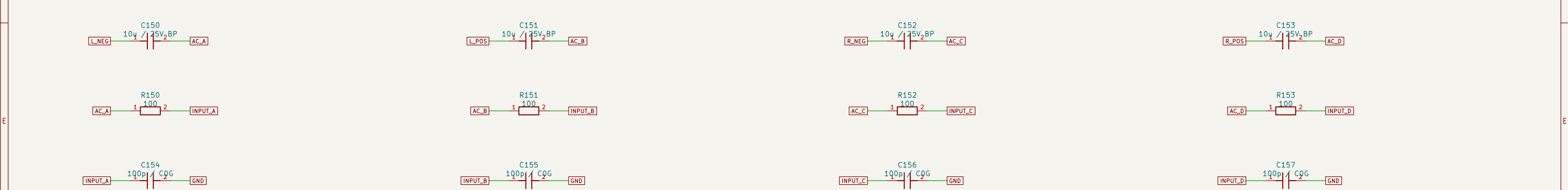
Rev: 0.2 REVIEW  
Id: 1/4

RCA INPUT / SINGLE-ENDED TO DIFFERENTIAL

NE5532 on 12V, VMID approx. 6V. Signal capacitors are bipolar/film.



AC coupling + input RF filters (COG shunt after 100 ohms)



Electrical draft – no PCB / footprints / hardware validation

Sheet: /input/  
File: input.kicad\_sch

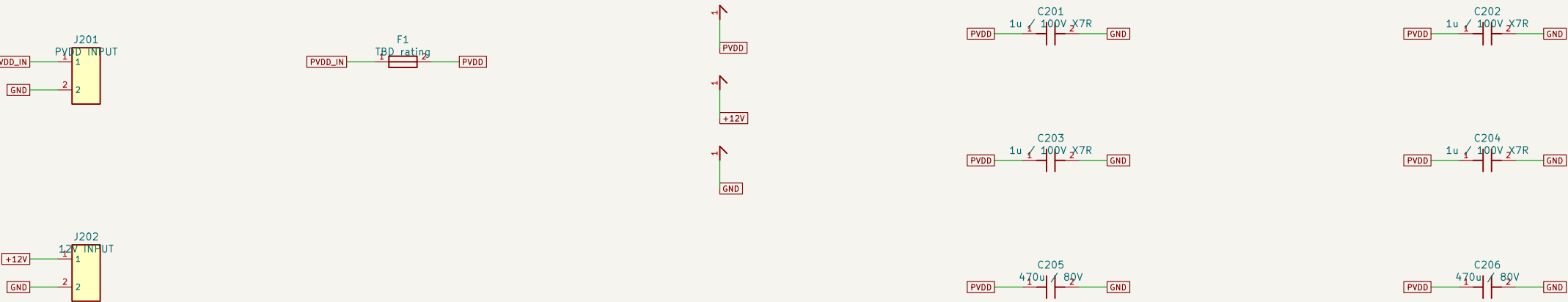
**Title: TPA3255 V0.2 / input**

Size: A3 Date: 2026-09-10  
KiCad E.D.A. 10.0.6

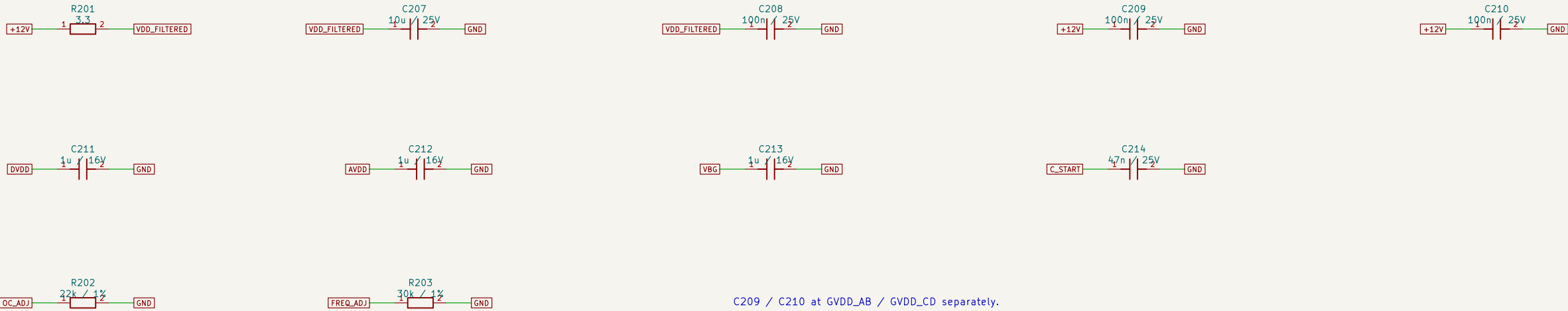
Rev: 0.2 REVIEW  
Id: 2/4

EXTERNAL SUPPLIES / LOCAL DECOUPLING / MANUAL RESET

Bench version: current-limited DC sources; no mains circuit or automatic power supervisor.



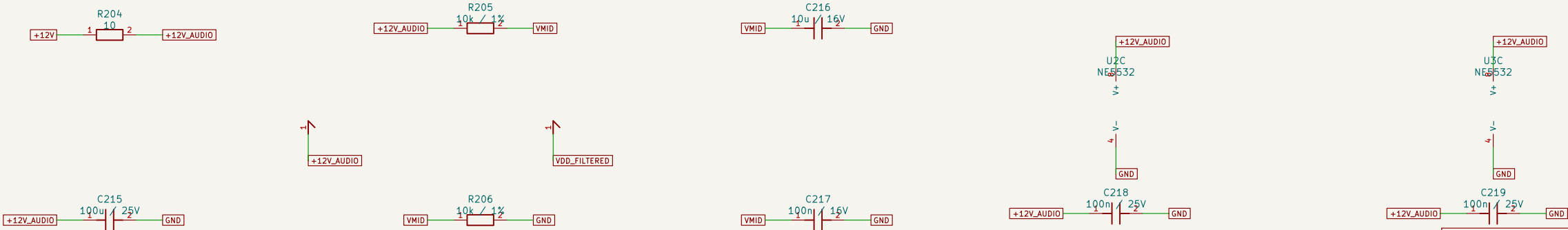
Place 2x MLCC + 1x bulk at EACH PVDD group.



C209 / C210 at GVDD\_AB / GVDD\_CD separately.

DVDD / AVDD / VBG: internal outputs, bypass only.

INPUT SUPPLY / VMID



Electrical draft – no PCB / footprints / hardware validation

Sheet: /power-control/  
File: power-control.kicad\_sch

**Title: TPA3255 V0.2 / power-control**

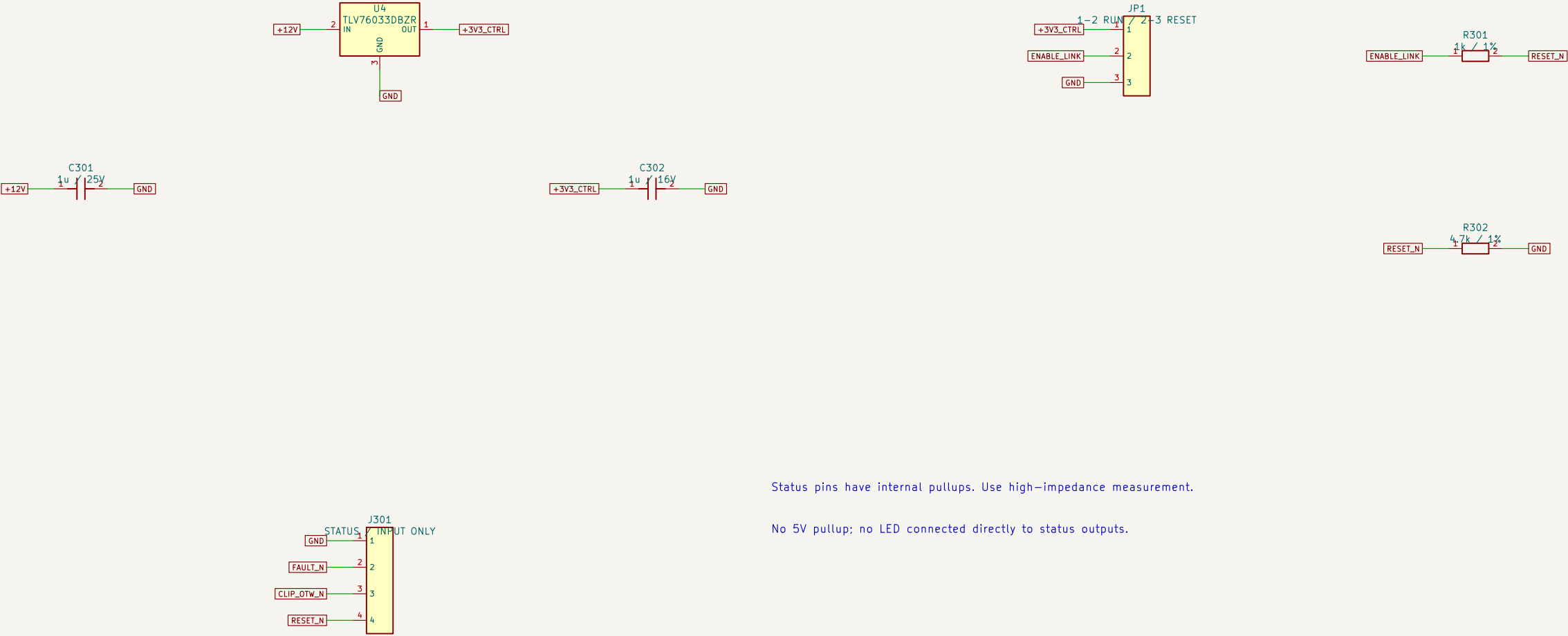
Size: A3  
KiCad E.D.A. 10.0.6

Date: 2026-09-10

Rev: 0.2 REVIEW  
Id: 3/4

MANUAL BENCH ENABLE / STATUS

Default: jumper absent => RESET low. This is not an automatic brownout supervisor.



Status pins have internal pullups. Use high-impedance measurement.

No 5V pullup; no LED connected directly to status outputs.

1. Keep JP1 open or 2-3 shorted. Connect only dummy loads.
2. Apply 12V and PVDD. Confirm rails and wait >= 250ms after PVDD.
3. Move JP1 to 1-2 to run. Start with small input and low PVDD.
4. Return JP1 to 2-3 before switching power off.

Do not use with speakers until automatic supply supervision and shutdown are designed.

|                                                              |                  |                 |
|--------------------------------------------------------------|------------------|-----------------|
| Electrical draft – no PCB / footprints / hardware validation |                  |                 |
| Sheet: /reset/<br>File: reset.kicad_sch                      |                  |                 |
| Title: TPA3255 V0.2 / reset                                  |                  |                 |
| Size: A3                                                     | Date: 2026-09-10 | Rev: 0.2 REVIEW |
| KiCad E.D.A. 10.0.6                                          |                  | Id: 4/4         |